

union : union stack set vertically, include duplicates (slower performance)
union all : removes duplicates (faster).

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COMMON TABLE EXPRESSION

A CTE is a temporary result set in SQL, that you can reference within a single query, created using a WITH clause

- break complex queries into smaller steps
- Improve readability
- Make transformations easier to understand
- avoid deeply nested queries
- can create multiple intermediate result sets
- useful with joins, aggregations, filtering and transformation.

Syntax:

```
with cte_name AS C
  select ...
)
select ...
from cte_name ;
```

```
with
↓
Define CTE
↓
Execute CTE Query
↓
create temporary result
↓
main query uses CTE
↓
Statement finishes
↓
CTE is gone
```

1. find department whose average salary is greater than 50000 & then show their employees. (with CTE)

with dept_avg as (

select department)

avg(salary) as avg_salary

from employees

group by department

)

select *

from dept_avg

where avg_salary > 50000;

when we want multiple CTE

with dept_avg as (

select department

avg(salary) as avg_salary

from employees

group by department

),

high_avg_depts as (

select *

from dept_avg

where avg_salary > 50000

)

select *

from high_avg_department

we can also use
having instead

with dept_salary as (

select dept,

avg(salary) as avg_salary

from employees

group by department

having avg(salary) > 50000

)

select * from dept_salary